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CNET 5050: Assignment 2

Assigned: 10/19/2026
Due: 10/29/2026

Introduction.
In Part 1 of this assignment, you will study the BA Model, Link selection model and the configuration model. Submit your written solutions as a PDF prepared using LaTeX or Microsoft Word, along with the source code used for your analysis. Ensure that your code is readable and appropriately commented. If you use generative AI tools such as ChatGPT, identify the tasks for which they were used and include the relevant prompts and generated outputs. In Part 2, you will develop your ability to critically read and present network science research papers.

Part 1

Before beginning Part 1, review Advanced Topic 3.A, "Power Laws," and Advanced Topic 3.B, "Plotting Power Laws," in the textbook: `https://networksciencebook.com/chapter/4#advanced-a`.

I. **Barabási–Albert Networks** **[20 points]**

Write a program that generates a Barabási–Albert network with $N = 10^4$ nodes and attachment parameter $m = 4$. Begin with a fully connected network of four nodes. Then add nodes one at a time, connecting each new node to m distinct existing nodes selected preferentially in proportion to their degree. Implement the model yourself; do not use a built-in Barabási–Albert network generator. Include your source code with your submission. Record snapshots of the growing network when it reaches $N = 10^2$, 10^3, and 10^4 nodes.

(a) Visualize one realization of the BA network with $N = 10^3$ nodes using a network visualization method of your choice.

(b) Generate $N_G = 100$ independent graph realizations. For each realization, record the degree of every node at each snapshot size, $N = 10^2$, 10^3, and 10^4. For each snapshot size N, pool the degree observations across all realizations, giving $N_G \times N$ observations. Plot the log-binned empirical degree distribution $\mathbb{P}(k)$ on logarithmic axes for each network size and compare it with the theoretical asymptotic degree distribution of the BA model derived in class,

$$\mathbb{P}(k) = \frac{2m(m+1)}{k(k+1)(k+2)} \approx 2m^2 k^{-3}. \tag{1}$$

This expression applies for $k \geq m = 4$. Clearly label the axes and distinguish the three network sizes in the legend. Refer to the lecture slides or the textbook on how to present the plots. What is the nature of the degree distribution in the BA model?

(c) Plot the complementary cumulative degree distributions, $\mathbb{P}(K \geq k)$, for the three network sizes together on the same axes. Clearly label the axes and legend.

(d) It is known that the average local clustering coefficient $\langle C \rangle$ in the BA model scales as $\langle C \rangle \sim (\ln N)^2/N$ (see Figure 5.19 and Advanced Topic 4.C). Generate $N_G = 100$ independent graph realizations of size $N = 10^4$. Measure the average clustering coefficient at several snapshots as each network grows, average the measurements across the N_G graphs, and plot the result as a function of network size N. Compare it with the theoretical scaling.

(e) Following Figure 5.6a, track the degree over time of one node from the initial four-node network and of nodes added at times $t = 10$, 100, 1000, and 5000. Define t as the number of node additions after the initial network. Plot the degree of each tracked node, averaged over

$N_G = 100$ graphs, as a function of t from its introduction through the end of the simulation. Compare the results with the theoretical scaling derived in class,

$$k_i(t) = m\sqrt{\frac{t}{t_i}}. \tag{2}$$

II. **The Friendship Paradox** **[30 points]**

The *link selection model* (see Chapter 5) is a simple local mechanism that generates a scale-free network without an explicit preferential-attachment rule. The model is defined as follows:

- *Growth*: At each time step, add a new node to the network.
- *Link selection*: Select a link uniformly at random, then connect the new node to one of its endpoints, chosen uniformly at random.

(a) Write a program that generates a network from the link selection model described above. Initialize the network with two nodes connected by an edge.

(b) Generate $N_G = 100$ independent graph realizations of size $N = 10^4$. Pool the degree observations across all realizations, giving $N_G \times N$ observations. Plot the log-binned empirical degree distribution $\mathbb{P}(k)$ on logarithmic axes for this network size. Compare the empirical distribution with Eq. (1) using $m = 1$. Explain how the link selection model creates an *effective* preferential-attachment mechanism by giving the form of $\Pi_i = \Pi(k_i)$, the probability that the incoming node connects to an existing node i of degree k_i.

(c) The *configuration model* is a paradigmatic "equilibrium" random network model and is often used as a *null model* to randomize connections while preserving a specified degree sequence and isolate the effects of degree heterogeneity. Let $\mathbb{P}(k)$ denote the degree distribution, where k is an integer. Show that the probability that a node at the end of a uniformly selected edge has degree k is $\mathcal{Q}(k) = k\mathbb{P}(k)/\langle k\rangle$, where $\langle k\rangle = \sum_k k\mathbb{P}(k)$ is the average degree.

(d) Using the form of $\mathcal{Q}(k)$, derive the expected degree $\langle k_{\text{neigh}}\rangle$ of a neighbor in the configuration model in terms of the first and second moments of $\mathbb{P}(k)$.

(e) Let $\mathbb{P}(k) = Ck^{-\gamma}$ for integer degrees $k_{\min} \le k \le k_{\max}$. Determine the normalization constant C and the first and second moments of this distribution in terms of γ, $k_{\min}$, and $k_{\max}$.

(f) For a network with $N = 10^4$, $\gamma = 2.3$, $k_{\min} = 1$, and $k_{\max} = 1000$, calculate the mean degree of a neighbor $\langle k_{\text{neigh}}\rangle$ and the network's mean degree $\langle k\rangle$. Use the discrete truncated power-law distribution specified above, and compare the two values.

(g) Explain the friendship paradox in light of your result: why do the friends of a randomly selected node have a higher mean degree than the nodes themselves?

Part 2 **[50 marks]**

Each student needs to choose one paper out of the following and present.

Network Structure, Robustness, and Centrality

- Albert, R., Jeong, H., & Barabási, A.-L. (2000). Error and attack tolerance of complex networks. *Nature, 406*(6794), 378–382. `https://doi.org/10.1038/35019019`
- Milo, R., Shen-Orr, S., Itzkovitz, S., Kashtan, N., Chklovskii, D., & Alon, U. (2002). Network motifs: Simple building blocks of complex networks. *Science, 298*(5594), 824–827. `https://doi.org/10.1126/science.298.5594.824`
- Page, L., Brin, S., Motwani, R., & Winograd, T. (1999). *The PageRank citation ranking: Bringing order to the web* (Technical Report No. 1999-66). Stanford InfoLab. `http://ilpubs.stanford.edu:8090/422/`

Epidemics and Information Diffusion

- Brockmann, D., & Helbing, D. (2013). The hidden geometry of complex, network-driven contagion phenomena. *Science, 342*(6164), 1337–1342. https://doi.org/10.1126/science.1245200

- Iribarren, J. L., & Moro, E. (2009). Impact of human activity patterns on the dynamics of information diffusion. *Physical Review Letters, 103*(3), Article 038702. https://doi.org/10.1103/PhysRevLett.103.038702

- Pastor-Satorras, R., & Vespignani, A. (2001). Epidemic spreading in scale-free networks. *Physical Review Letters, 86*(14), 3200–3203. https://doi.org/10.1103/PhysRevLett.86.3200

Community Detection and Statistical Inference

- Fortunato, S., & Barthélemy, M. (2007). Resolution limit in community detection. *Proceedings of the National Academy of Sciences, 104*(1), 36–41. https://doi.org/10.1073/pnas.0605965104

- Newman, M. E. J. (2006). Modularity and community structure in networks. *Proceedings of the National Academy of Sciences, 103*(23), 8577–8582. https://doi.org/10.1073/pnas.0601602103

- Peixoto, T. P. (2017). Nonparametric Bayesian inference of the microcanonical stochastic block model. *Physical Review E, 95*(1), Article 012317. https://doi.org/10.1103/PhysRevE.95.012317

Social Networks, Politics, and Misinformation

- Adamic, L. A., & Glance, N. (2005). The political blogosphere and the 2004 U.S. election: Divided they blog. In J. Adibi, M. Grobelnik, D. Mladenic, & P. Pantel (Eds.), *Proceedings of the 3rd International Workshop on Link Discovery* (pp. 36–43). Association for Computing Machinery. https://doi.org/10.1145/1134271.1134277

- Grinberg, N., Joseph, K., Friedland, L., Swire-Thompson, B., & Lazer, D. (2019). Fake news on Twitter during the 2016 U.S. presidential election. *Science, 363*(6425), 374–378. https://doi.org/10.1126/science.aau2706

- Lazer, D., Rubineau, B., Chetkovich, C., Katz, N., & Neblo, M. (2010). The coevolution of networks and political attitudes. *Political Communication, 27*(3), 248–274. https://doi.org/10.1080/10584609.2010.500187

- Traud, A. L., Mucha, P. J., & Porter, M. A. (2012). Social structure of Facebook networks. *Physica A: Statistical Mechanics and its Applications, 391*(16), 4165–4180. https://doi.org/10.1016/j.physa.2011.12.021

Network Neuroscience

- Achard, S., Salvador, R., Whitcher, B., Suckling, J., & Bullmore, E. (2006). A resilient, low-frequency, small-world human brain functional network with highly connected association cortical hubs. *Journal of Neuroscience, 26*(1), 63–72. https://doi.org/10.1523/JNEUROSCI.3874-05.2006

- Bullmore, E., & Sporns, O. (2009). Complex brain networks: Graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience, 10*(3), 186–198. https://doi.org/10.1038/nrn2575

- Hagmann, P., Cammoun, L., Gigandet, X., Meuli, R., Honey, C. J., Wedeen, V. J., & Sporns, O. (2008). Mapping the structural core of human cerebral cortex. *PLOS Biology, 6*(7), Article e159. https://doi.org/10.1371/journal.pbio.0060159

Network Medicine

- Goh, K.-I., Cusick, M. E., Valle, D., Childs, B., Vidal, M., & Barabási, A.-L. (2007). The human disease network. *Proceedings of the National Academy of Sciences, 104*(21), 8685–8690. https://doi.org/10.1073/pnas.0701361104

Presentation Guidelines

Total duration: 15–20 minutes

- 12–15 minutes for the presentation
- 3–5 minutes for questions and discussion

Suggested Organization

- Introduction and context (2 minutes)
- Network construction and data (2–3 minutes)
- Methods and analysis (3–4 minutes)
- Results and interpretation (3–4 minutes)
- Critical discussion and implications (2 minutes)

Guiding Questions

System and Network Construction

- What real-world system or phenomenon does the paper study?
- How is the network constructed? Clearly identify the nodes, edges, and any edge weights or directions.
- What data are used to construct the network?

Measures and Methods

- Which network measures, models, or algorithms are applied, such as degree, assortativity, centrality, community detection, or motif analysis?
- Are null models or randomization tests used? If so, what purpose do they serve?
- What aspects of the system's structure do the chosen methods capture?

Findings and Interpretation

- What are the paper's main findings and structural patterns?
- How do the authors interpret these patterns in the context of the underlying system?

Critical Reflection

- What assumptions or limitations affect the network representation, data, methods, or interpretation?
- What alternative methods or measures could strengthen or extend the analysis?

Broader Impact

- What is the main takeaway about the real-world system being studied?
- How does the paper advance our understanding of this system or demonstrate the value of a network-science approach?